

CROSS-BASE CORRELATIONS OF ARTIN PRIMES: ENTANGLEMENT, EXCLUSION, AND THE STRUCTURE OF $p - 1$

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ABSTRACT. Say a prime p is *Artin base* a if a is a primitive root modulo p . This paper studies the joint behaviour of Artin status *across bases* at the same prime. We first record a deterministic exclusion law: if $\text{sqf}(ab)$ is the squarefree part of ab and c is any base with $\text{sqf}(c) = \text{sqf}(ab)$, then no prime is simultaneously Artin for a , b , and c — for instance, no prime has 2, 5, and 10 all as primitive roots. The proof is three lines from the multiplicativity of quadratic characters. We then measure, for all 66 pairs from the twelve bases $\{2, 3, 5, 6, 7, 10, 11, 13, 15, 17, 21, 29\}$ and all 50,847,532 primes up to 10^9 , the same-prime correlation $\phi(a, b)$ between the indicator variables of Artin status. Three empirical findings emerge. First, the correlation is positive for 64 of the 66 pairs (mean $\phi = 0.0352$, individual $|z|$ up to 476); the only negative pairs, (2, 10) and (3, 15), are precisely pairs whose product is (a square times) another base — the entangled configurations. Second, conditioning on the number of prime factors $\omega(p - 1)$ removes only 55% of the mean correlation, but conditioning on a finer signature of $p - 1$ — the 2-adic valuation, the divisibility of $p - 1$ by each prime up to 13, and the number of larger factors — removes 95%: the cross-base correlation is, almost entirely, a reflection of the fact that all bases see the same factorisation of $p - 1$. Third, the residual that survives fine conditioning is concentrated exactly on the multiplicatively dependent pairs (mean residual $|\phi| \approx 0.005$ within dependent pairs, versus 0.001 for multiplicatively independent pairs), and is removed by additionally conditioning on the quadratic-residue status of the bases — identifying it as character-level entanglement, the same mechanism as the deterministic exclusion law. The measurements are consistent with the Galois-theoretic account of simultaneous primitive roots going back to Matthews and to Lenstra, and give it a quantitative, channel-by-channel verification at scale.

1. INTRODUCTION

Artin’s conjecture on primitive roots concerns one base at a time: for non-square $a \neq -1$, the set

$$A_a = \{p \text{ prime} : a \text{ is a primitive root mod } p\}$$

should have density given by a rational multiple of Artin’s constant, as proved under GRH by Hooley [1]. The joint distribution for *several* bases at the same prime was treated by Matthews [4], who computed (under GRH) the density of primes for which several prescribed integers are simultaneously primitive roots, and by Lenstra [2], whose Galois-theoretic framework subsumes such questions; see Moree’s survey [5]. The qualitative message of that theory is that the events $p \in A_a$ and $p \in A_b$ are *not* independent, and that the deviation from independence is governed by the interaction of the Kummer–Galois extensions attached to a and b — in modern language, by *entanglement* between them.

This paper is the third in a series studying correlations of Artin status computationally at the scale of 10^9 . The first paper measured the correlation between the Artin statuses (base 10) of *consecutive* primes and decomposed it into residue channels; the second proved a classification

Date: August 17, 2026.

2020 Mathematics Subject Classification. Primary 11A07; Secondary 11N05, 11N13, 11Y16, 11Y60.

Key words and phrases. Artin’s conjecture, primitive roots, simultaneous primitive roots, entanglement, quadratic characters, Kummer theory, computational number theory.

of deterministic exclusion laws for consecutive Artin primes in arbitrary bases and identified the conductor of the quadratic character χ_a as the parameter governing the statistical correlation. Here we complete the picture along the remaining axis: *fix the prime, vary the base*.

Two questions organise the paper.

(Q1) *What is deterministically forbidden?* For consecutive primes, the answer (paper two of the series) was a set of gap classes on which the quadratic character is forced to flip. For a single prime and several bases, the answer is an exclusion law inside *multiplicative triples*:

Theorem 1 (Same-prime exclusion law; see Theorem 2). *Let a, b be non-square positive integers and let c be any non-square positive integer with $\text{sqf}(c) = \text{sqf}(ab)$. Then no prime $p \nmid 2abc$ is simultaneously Artin base a , Artin base b , and Artin base c .*

For example, no prime has all of 2, 5, 10 as primitive roots; no prime has all of 3, 5, 15; and, since sqf is all that matters, no prime has all of 2, 5, 40. Like the gap exclusion law of the second paper, the statement is elementary — it needs only the multiplicativity of the quadratic character and the fact that a quadratic residue is not a primitive root — and, like that law, we have not found it stated explicitly in the literature, although it is certainly implicit in the density computations of [4, 2], where the entangled character relation $\chi_a \chi_b = \chi_{ab}$ is exactly what produces degenerate factors.

(Q2) *What does the joint distribution actually look like, and what explains it?* Here the data speak. We compute the exact 2×2 contingency table of $(\mathbf{1}[p \in A_a], \mathbf{1}[p \in A_b])$ over all 50,847,532 primes $p \leq 10^9$, for all 66 pairs from the base set

$$\{2, 3, 5, 6, 7, 10, 11, 13, 15, 17, 21, 29\},$$

chosen to contain many multiplicative triples (15 of the 66 pairs lie in a triple $\text{sqf}(ab) = \text{sqf}(c)$ within the set) alongside multiplicatively independent pairs. The findings, developed in Sections 4–6:

- (1) *Positivity.* $\phi(a, b) > 0$ for 64 of 66 pairs, with mean 0.0352 and z -scores up to 476: knowing that a is a primitive root mod p makes it *more* likely that b is. The largest correlations all involve base 5 (e.g. $\phi(5, 13) = 0.0667$: $P(A_{13} \mid A_5) = 0.416$ against $P(A_{13} \mid \neg A_5) = 0.350$).
- (2) *The exceptions are the entangled pairs.* The only two negative correlations are $\phi(2, 10) = -0.0302$ and $\phi(3, 15) = -0.0303$ — precisely the pairs (a, b) in our set with $b = a \cdot c$ for another base c of the set (up to squares). Multiplicative dependence reverses the sign of the association.
- (3) *The positive correlation is $p - 1$ structure, not entanglement.* Conditioning on $\omega(p - 1)$ removes 55% of the mean correlation; conditioning on a fine signature of $p - 1$ — namely $(\min(v_2(p - 1), 4), \text{the set of primes } q \leq 13 \text{ dividing } p - 1, \min(\omega_{>13}(p - 1), 3))$ — removes 95%, leaving a mean residual of 0.0019. All bases are tested against the same integer $p - 1$; a prime whose $p - 1$ is “primitive-root friendly” (few small factors, small 2-part) is friendlier for *every* base simultaneously. This common-cause structure, not any interaction between the bases, is the bulk of the correlation.
- (4) *What survives is entanglement, and it is exactly localised.* After fine conditioning, the residual correlation among multiplicatively *dependent* pairs (mean $|\phi_{\text{sig}}| \approx 0.005$, and up to -0.026 for $(5, 15)$, -0.012 for $(5, 10)$) is five times that of independent pairs (0.001); and in a separate run additionally conditioning on the quadratic-residue status of the bases modulo p flattens the dependent pairs to the level of the independent ones. The residual is thus identified, empirically, as the quadratic-character entanglement $\chi_a \chi_b = \chi_{\text{sqf}(ab)}$ — the statistical shadow of Theorem 2.

The theme of the series persists: at every level, the joint behaviour of Artin primes decomposes into a small deterministic core (here, the triple exclusion law) plus a statistical component with an identifiable arithmetic carrier (here, the shared factorisation of $p - 1$, with entanglement as a measurable second-order correction).

2. THE SAME-PRIME EXCLUSION LAW

Throughout, for a non-square positive integer a we write $d_a = \text{sqf}(a)$ for its squarefree part and $\chi_a = \left(\frac{d_a}{\cdot}\right)$ for the quadratic character attached to $\mathbb{Q}(\sqrt{a})$; thus $\chi_a(p) = +1$ exactly when a is a quadratic residue modulo p (for $p \nmid 2a$), and

$$p \in A_a \implies \chi_a(p) = -1, \quad (1)$$

since a quadratic residue generates a subgroup of index at least 2 in $(\mathbb{Z}/p\mathbb{Z})^\times$.

Theorem 2 (Triple exclusion). *Let a, b, c be non-square positive integers with $\text{sqf}(c) = \text{sqf}(ab)$, and let $p \nmid 2abc$ be prime. Then p is not simultaneously Artin base a , Artin base b , and Artin base c .*

Proof. For any prime $p \nmid 2ab$ one has, by complete multiplicativity of the Legendre symbol in the numerator, $\chi_a(p)\chi_b(p) = \left(\frac{d_a d_b}{p}\right) = \left(\frac{\text{sqf}(ab)}{p}\right) = \chi_c(p)$, since $d_a d_b$ and $\text{sqf}(ab)$ differ by a square. If p were Artin for all three bases, then by (1) all three character values would be -1 , giving $(-1)(-1) = -1$, a contradiction. $\square$

Corollary 3. *Within any multiplicative triple — for instance $(2, 5, 10)$, $(3, 5, 15)$, $(3, 7, 21)$, $(2, 3, 6)$ — a prime can be Artin for at most two of the three bases.*

Remark 4. The exclusion is sharp: primes Artin for exactly two of $(2, 5, 10)$ exist in abundance. The smallest is $p = 7$, where 5 and 10 are both primitive roots (each of order 6) while 2 has order 3; likewise $p = 23$, where 5 and 10 are primitive roots and 2 is a quadratic residue of order 11. More systematically, in our data the pair $(2, 10)$ has $P(A_{10} \mid A_2) = 0.355$: two-of-three is common, three-of-three is impossible.

Remark 5. Theorem 2 is the same-prime analogue of the gap exclusion law for consecutive primes proved in the second paper of this series: both are instances of a parity obstruction in the group generated by the quadratic characters involved. There the two characters were χ_a evaluated at p and at $p + g$, and the obstruction was arranged by the shift g ; here the characters belong to different bases at the same prime, and the obstruction is arranged by multiplicative dependence. The Galois-theoretic density computations of Matthews [4] and Lenstra [2] encode this relation implicitly (the degree of the compositum of the relevant Kummer extensions drops); the pointwise statement seems worth recording.

3. DATA AND COMPUTATION

By a segmented sieve we enumerated all primes $p \leq 10^9$ (50,847,534 primes, of which 50,847,532 satisfy $p \nmid 2 \cdot (\text{base})$ for all twelve bases and enter the analysis). For each prime we factored $p - 1$ once by trial division and computed the Artin indicator for each of the twelve bases by the standard criterion ($a^{(p-1)/\ell} \not\equiv 1 \pmod{p}$ for every prime $\ell \mid p - 1$, in 128-bit arithmetic). One streaming pass accumulates, for each of the 66 unordered base pairs, the 2×2 contingency table of joint Artin status — globally, and stratified by (i) $\omega(p - 1)$ and (ii) the fine signature

$$\text{sig}(p) = \left(\min(v_2(p - 1), 4), \{q \leq 13 : q \mid p - 1\}, \min(\omega_{>13}(p - 1), 3) \right),$$

TABLE 1. Same-prime cross-base Artin correlation at $x = 10^9$: summary and extremes. “Dependent” means the pair lies in a multiplicative triple within the base set ($\text{sqf}(ab) = \text{sqf}(c)$).

	pairs	mean ϕ
all pairs	66	+0.0352
multiplicatively independent	51	+0.0363
multiplicatively dependent	15	+0.0314
largest: (5, 13)		+0.0667
smallest: (3, 15)		−0.0303
second smallest: (2, 10)		−0.0302

TABLE 2. The ten largest $|\phi|$ at $x = 10^9$, with the values after conditioning on $\omega(p-1)$ and on the fine signature $\text{sig}(p)$ (pair-weighted mean of within-cell ϕ ; cells with fewer than 200 primes excluded). “Dep.” marks multiplicatively dependent pairs.

a	b	ϕ	$\phi \mid \omega$	$\phi \mid \text{sig}$	Dep.
5	13	+0.0667	+0.0200	+0.0003	
5	17	+0.0649	+0.0218	+0.0025	
5	29	+0.0631	+0.0246	+0.0007	
5	10	+0.0624	+0.0285	−0.0125	✓
5	6	+0.0623	+0.0292	+0.0005	
5	11	+0.0623	+0.0300	+0.0004	
5	15	+0.0622	+0.0297	−0.0255	✓
3	5	+0.0622	+0.0284	+0.0015	✓
5	7	+0.0622	+0.0297	+0.0002	
2	5	+0.0621	+0.0336	+0.0004	✓

a partition of the primes into at most $5 \cdot 32 \cdot 4 = 640$ cells. The signature records exactly the information about $p-1$ that the Artin criterion consults for small bases: the 2-part and the small odd prime divisors. A separate run to 3×10^8 additionally stratified by the vector of quadratic-residue statuses $(\chi_a(p))_a$ of the twelve bases. The whole computation runs in under two hours on a single core of a commodity machine (2-core Intel Xeon E5-2673 v4, 8 GB RAM); all counts are exact.

As validation, the marginal Artin densities matched the Hooley densities to four decimal places for every base — e.g. 0.373993 for the bases with squarefree part $\not\equiv 1 \pmod{4}$, against Artin’s constant 0.373956, and 0.393642 for base 5, whose density carries the classical correction factor $C \cdot \frac{20}{19}$ for $d \equiv 1 \pmod{4}$ (see [1, 5]).

4. THE CROSS-BASE CORRELATION

For each pair write $\phi(a, b)$ for the Pearson correlation of the indicator variables (the ϕ coefficient of the 2×2 table). At $N \approx 5.08 \times 10^7$ primes, the standard error of ϕ under independence is $N^{-1/2} \approx 0.00014$; as in the previous papers of the series, the enormous resulting z -scores certify only that sampling noise is negligible, and the relevant quantity is the effect size.

TABLE 3. Mean cross-base correlation after conditioning, all 66 pairs at $x = 10^9$.

conditioning	mean ϕ	removed
none	0.0352	—
$\omega(p-1)$	0.0158	55%
$\text{sig}(p)$: $(v_2, \{q \leq 13\}, \omega_{>13})$	0.0019	95%

Three observations, in increasing order of depth.

Observation 6 (Positivity). $\phi(a, b) > 0$ for 64 of the 66 pairs. Artin statuses of different bases at the same prime *attract*.

The sign is the opposite of the consecutive-prime correlation measured in the first two papers of the series (where $\delta < 0$ uniformly), and the mechanism is transparent: there the two primes had *different* (and LOS-anticorrelated) values of $p-1$; here every base is tested against the *same* $p-1$. If $p-1$ has few small prime factors and a small 2-part, the primitive-root condition is easier for every base at once. Formally, the Artin indicators share the common cause $\text{sig}(p)$.

Observation 7 (The negative pairs are the entangled ones). The only pairs with $\phi < 0$ are $(2, 10)$ and $(3, 15)$: in each, the product of the two bases is, up to a square, the third base of a multiplicative triple contained in the set $(2 \cdot 10 = 4 \cdot 5, 3 \cdot 15 = 9 \cdot 5)$. For $(2, 10)$: $P(A_{10} \mid A_2) = 0.3551$ versus $P(A_{10} \mid \neg A_2) = 0.3853$.

The sign reversal has a character-theoretic origin. For a dependent pair, $\chi_a \chi_b = \chi_5$ (in both our examples), so on the half of primes with $\chi_5(p) = +1$ the values $\chi_a(p)$ and $\chi_b(p)$ are forced *equal* — fine for joint Artin status — but on the half with $\chi_5(p) = -1$ they are forced *opposite*, and then at most one of the two bases can be Artin: a pointwise exclusion on half the primes, diluted into the observed negative correlation. This is Theorem 2 operating statistically through the third character.

5. DECOMPOSITION: THE CORRELATION IS THE STRUCTURE OF $p-1$

Conditioning on $\omega(p-1)$ alone — the crude size of the factorisation — removes barely half of the correlation. This is worth emphasising because ω is the statistic through which the consecutive-prime papers routed the mechanism, and it is genuinely insufficient here: two primes with the same $\omega(p-1)$ can present very different Artin conditions (it matters *which* primes divide $p-1$, and with what 2-adic valuation, not how many). The fine signature — 2-adic valuation capped at 4, the exact set of primes ≤ 13 dividing $p-1$, and the count of larger prime factors capped at 3 — is the information the Artin criterion actually consults for small bases, and conditioning on it removes 95% of the mean correlation (Table 3).

We regard this as the main statistical finding: *cross-base Artin correlation at a single prime is, to first order, not an interaction between the bases at all*. It is a common-cause effect of the shared integer $p-1$. Any model that treats A_a and A_b as conditionally independent given the small-prime structure of $p-1$ reproduces 95% of the observed association.

6. THE RESIDUAL IS ENTANGLEMENT

The 5% that survives is not noise, and it is not uniformly distributed: it sits precisely on the multiplicatively dependent pairs.

Two facts pin down the mechanism. First, after fine conditioning the dependent pairs retain residuals up to -0.026 (for $(5, 15)$) and -0.012 (for $(5, 10)$) — large negative values, of the sign

TABLE 4. Residual correlation after conditioning on $\text{sig}(p)$, split by multiplicative dependence, at $x = 10^9$; and the effect of additionally conditioning on the quadratic-residue vector (run to 3×10^8).

group	mean ϕ sig	mean ϕ sig + QR
multiplicatively dependent pairs	+0.0049	+0.0042
of which sharing a common factor	−0.0052*	+0.0042
multiplicatively independent pairs	+0.0010	+0.0050

*Signed values; the extreme cells are (5, 15) at −0.0255 and (5, 10) at −0.0125 (see Table 2).

predicted by the character relation of Observation 7 — whereas the 51 independent pairs sit at a mean of +0.001. Second, in the run to 3×10^8 that additionally conditions on the vector of quadratic-residue statuses $(\chi_a(p))_a$, the negative residuals of the dependent pairs vanish: every group lands in the narrow band 0.003–0.006 (Table 4), consistent with the small positive floor attributable to higher-order (cubic and quartic) conditions shared through $p - 1$ beyond our signature. Conditioning on the quadratic characters is exactly what removes character-level entanglement; that it flattens precisely the pairs with $\chi_a \chi_b = \chi_c$, and only those, identifies the residual as the statistical trace of the entangled relation — the same object that makes Theorem 2 true.

The picture that emerges is cleanly layered:

$$\underbrace{\text{common cause } p - 1}_{\approx 95\%} + \underbrace{\text{quadratic entanglement } \chi_a \chi_b = \chi_{\text{sqf}(ab)}}_{\text{residual, sign-definite, localised}} + \underbrace{\text{higher-order sharing}}_{\text{small positive floor}}.$$

7. DISCUSSION

Relation to the theory. Nothing in our findings contradicts the classical account; rather, the measurements give it an explicit quantitative form that we have not seen computed. The densities of Matthews [4] for simultaneous primitive roots, and Lenstra’s general framework [2], predict the *joint* densities of (A_a, A_b) under GRH, including the entanglement corrections; our contingency tables are, in effect, a 50-million-prime test of those predictions across 66 pairs simultaneously, and the layered decomposition (common cause, then entanglement, then higher order) is a statement about *where* in the Galois structure the observed correlation lives. It would be worthwhile to extract from [4] the predicted $\phi(a, b)$ for each of our pairs and compare cell by cell; we have not done so here, and flag it as the natural next step for a reader with the stomach for the relevant index computations.

Relation to the first two papers of the series. The trilogy now covers both axes of the joint distribution of Artin status. Along the prime axis (consecutive p_n, p_{n+1} , one base), correlation is negative, driven by the Lemke Oliver–Soundararajan bias [3] propagating through the two distinct integers $p_n - 1$ and $p_{n+1} - 1$, with deterministic exclusion classes in the gap. Along the base axis (one prime, two bases), correlation is positive, driven by the single shared integer $p - 1$, with deterministic exclusion inside multiplicative triples. In both cases the deterministic law is a parity statement about quadratic characters, and in both cases it is statistically dominated by a factorisation-of- $p - 1$ mechanism — diluted to invisibility in the global average there, absorbed almost entirely by conditioning here.

Future work. (1) A cell-by-cell comparison with the Matthews–Lenstra predicted joint densities, which would upgrade Section 6 from identification to verification. (2) The mixed

axis: ϕ between $A_a(p_n)$ and $A_b(p_{n+1})$ for $a \neq b$ — the data pipeline extends immediately, and the prediction from the present findings is a small negative correlation governed by the two conductors jointly. (3) Higher-order entanglement: triples of bases conditioned on quadratic data should expose the cubic layer ($\ell = 3$ Kummer conditions), with $\mathbb{Q}(\zeta_3)$ playing the role $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{2})$ played here.

ACKNOWLEDGEMENTS

The author thanks the developers of the open-source tools used in this work (GCC, Python, `sympy`, `matplotlib`).

Use of generative AI. The author used Anthropic’s Claude (models Claude Opus 4.5 and Claude Opus 5) as an assistant: for drafting and revising exposition, for writing and debugging the sieve and analysis programs, and for exploratory discussion. All computations were executed and verified on the author’s own hardware from source code the author has inspected; the proof of Theorem 2 has been checked by hand; every reference has been verified against the published record. The author is solely responsible for the content.

DECLARATION OF INTEREST

The author reports there are no competing interests to declare.

FUNDING

This research received no external funding.

DATA AVAILABILITY

All code and results are publicly available at <https://github.com/jpbald93/consecutive-artin>, including the single-pass multi-base programs (`crossbase.c`, `crossbase_fine.c`, `crossbase_qr.c`), the exact contingency tables underlying every value quoted here, and the analysis scripts. The full computation reproduces in under two hours on a single core.

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